



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Advanced)

FULL SYLLABUS

12-07-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE : [SCORE-II (ALL STAR BATCH)]

ANSWER KEY

PAPER-1

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4		
	A.	C	D	C	C		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	B,D	B,C	A,D	A,B,C,D	B,C,D	A,C
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	2.00	90.00	18.00	9.50	0.10	0.40
SECTION-II (ii)	Q.	7	8	9			
	A.	9	3	2			

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4		
	A.	A	B	B	C		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	A,B,C	B,D	B,D	A,B,C	A,C,D	B
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	10.00	4.00	4.00	6.00	6.00	12.00
SECTION-II (ii)	Q.	7	8	9			
	A.	8	7	3			

PART-3 : MATHEMATICS

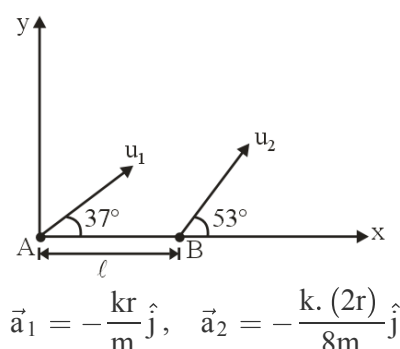
SECTION-I (i)	Q.	1	2	3	4		
	A.	D	C	B	A		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	A,B,C,D	A,C,D	A,B,C,D	A,C,D	A,B,D	B,C
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	8.00	-0.25	1.00	-1.00	1.00	1.00
SECTION-II (ii)	Q.	7	8	9			
	A.	6	2	6			

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (C)



$$= -4\hat{j} \quad \vec{a}_2 = -\frac{kr}{4m}\hat{j} = -\hat{j}$$

$$x_{AB} = u_{x(AB)} \times t$$

$$\Rightarrow \ell = \left(\frac{4u_1}{5} - \frac{3u_2}{5} \right) t \dots\dots\dots (1)$$

$$y_{AB} = u_{y(AB)} \cdot t + \frac{1}{2} a_{y(AB)} t^2$$

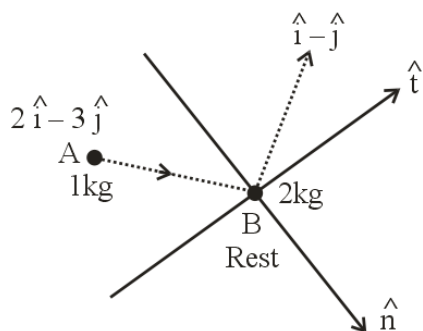
$$\Rightarrow 0 = \left(\frac{3u_1 - 4u_2}{5} \right) t + \frac{1}{2} (-3) t^2$$

$$\Rightarrow \left(\frac{3u_1 - 4u_2}{5} \right) = \frac{3t}{2}$$

$$\Rightarrow \left(\frac{3 \times 15 - 4 \times 5}{5} \right) = \frac{3t}{2} \Rightarrow t = \frac{10}{3}$$

$$\text{From (1)} \quad \ell = \left(\frac{4 \times 15 - 3 \times 5}{5} \right) \times \frac{10}{3} = 30$$

2. Ans (D)



$$A, \Delta \vec{p}_A = -\hat{i} + 4\hat{j}$$

$$B, \Delta \vec{p}_B = \hat{i} - 4\hat{j}$$

$$\hat{n} = \frac{\hat{i} - 4\hat{j}}{\sqrt{17}}$$

$$u_n = |\vec{u} \cdot \hat{n}| = \left| (2\hat{i} - 3\hat{j}) \cdot \frac{(\hat{i} - 4\hat{j})}{\sqrt{14}} \right|$$

$$= \frac{14}{\sqrt{17}}$$

$\therefore$ Max. K.E. loss

$$= \frac{1}{2} u V_r^2 = \frac{1}{2} \times \left(\frac{1 \times 2}{1+2} \right) \times \frac{196}{17}$$

$$= \frac{136}{\sqrt{17}}$$

3. Ans (C)

$$a = \frac{qE}{m} = \frac{q}{m} \cdot \frac{V}{d}$$

$$a = \frac{qV_0}{md} \left(1 - e^{-t/RC} \right)$$

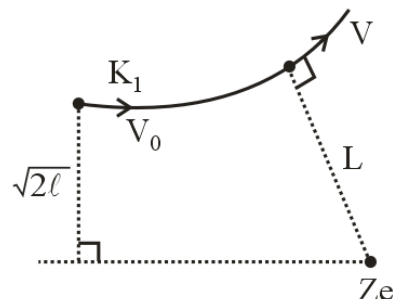
$$\therefore V = \int_0^{RC} a dt = \frac{qV_0}{md} \left(t - \frac{e^{-t/RC}}{-1/RC} \right)_0^{RC}$$

$$= \frac{qV_0}{md} \left[RC + \frac{RC}{e} - RC \right] = \frac{qV_0 RC}{mde}$$

4. Ans (C)

$$K_0 = \frac{ke \cdot Ze}{\ell} \Rightarrow \ell = \frac{kZe^2}{K_0}$$

From angular momentum conservation



$$\sqrt{2} m V_0 \ell = m v L$$

$$\Rightarrow V = \frac{\sqrt{2} V_0 \ell}{L} \quad \dots\dots (1)$$

From Energy conservation

$$K_0 = \frac{1}{2} m v^2 + \frac{KZe \cdot e}{L}$$

$$\Rightarrow K_0 = \frac{1}{2} m \left(\frac{\sqrt{2} V_0 \ell}{L} \right)^2 + \frac{KZe^2}{L}$$

$$\Rightarrow K_0 = \left(\frac{1}{2} m V_0^2 \right) \left(\frac{2\ell^2}{L^2} \right) + \frac{K_0 \ell}{L}$$

$$\Rightarrow K_0 = K_0 \left(\frac{2\ell^2}{L^2} \right) + K_0 \frac{\ell}{L}$$

$$\Rightarrow 2 \left(\frac{\ell}{L} \right)^2 + \left(\frac{\ell}{L} \right) - 1 = 0$$

$$\Rightarrow \frac{\ell}{L} = \frac{-1 + \sqrt{1+8}}{4} = \frac{1}{2}$$

$$\Rightarrow L = 2\ell$$

PART-1 : PHYSICS

SECTION-I (ii)

5. Ans (B,D)

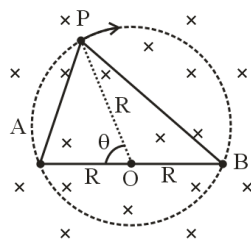
$$L.C. = \frac{P}{N} = \frac{2}{100} = 0.02 \text{ mm}$$

$$LSR = 4 \text{ mm}$$

$$CSR = 70 \times 0.02 - 1 = 0.4 \text{ mm}$$

$$\text{Thickness} = 4.4 \text{ mm}$$

6. Ans (B,C)

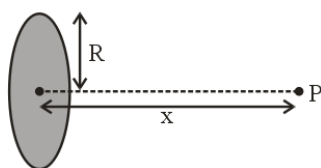


$$e = -\frac{d}{dt}(B \cdot \frac{1}{2} \cdot 2\pi R \cdot R \sin \theta)$$

$$e = -BR^2 \omega \cos \omega t$$

$$e = -BRV \cos\left(\frac{Vt}{R}\right)$$

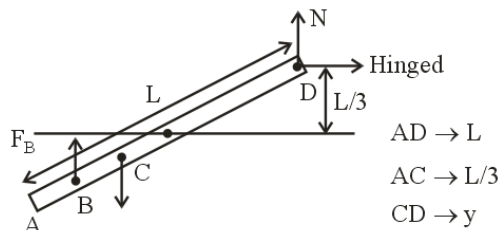
7. Ans (A,D)



$$B_{\text{on axis}} = \frac{\mu_0 \sigma \omega}{2} \left(\frac{2x^2 + R^2}{\sqrt{R^2 + x^2}} - 2x \right)$$

$$\text{near } 0, B_0 = \frac{1}{2} \mu_0 \sigma \omega R, B_0 \propto R$$

8. Ans (A,B,C,D)



For rotational equilibrium,

$$N \times O - w \times \frac{L}{2} + F_B \times \left(y + \frac{L-y}{2} \right) = 0$$

ρ - density of water.

$$w \frac{L}{2} = FB \left(y + \frac{L}{2} - \frac{y}{2} \right)$$

$$LA \frac{2}{3} \rho g \frac{L}{2} = (L-y) A \frac{3}{2} \rho g \left(\frac{y}{2} + \frac{L}{2} \right) = (L^2 - y^2) \frac{3}{4} A \rho g$$

$$\frac{4}{q} L^2 = L^2 - y^2, y^2 = L^2 - L^2 \frac{4}{q} = \frac{5L^2}{9}$$

$$\left[y = \pm \frac{\sqrt{5}L}{3} \right]$$

$$\text{weight of rod in air} = LA \frac{2}{3} \rho g$$

$$\text{Apparent weight of rod} = LA \frac{2}{3} \rho g - (L-y) A \frac{3}{2} \rho g$$

$$\text{inclination, } \sin \alpha = \frac{\frac{L}{3}}{y} = \frac{\frac{L}{3}}{\frac{\sqrt{5}L}{3}} = \frac{1}{\sqrt{5}}$$

9. Ans (B,C,D)

$$T_{RG} = 2000K$$

$$T_{WD} = 20,000K$$

$$\text{Radius } R_{WD} = 2000 \text{ Km}$$

$$V_{RG} = 10^{15} V_{WD}$$

$$R_{RG} = 10^5 R_{WD} = 2000 \times 10^5 \text{ KM}$$

$$\frac{P_{\text{ar.W.D}}}{R_{\text{av.RG}}} = \frac{T_{WD}^4}{T_{RG}^4} = \frac{(20,000)^4}{(2000)^4} = 10^4$$

$$P_{\text{Total RG}} = \sigma A_{RG} T_{RG}^4$$

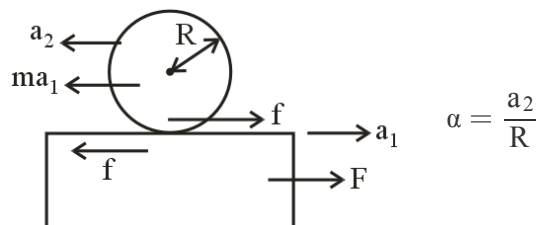
$$= 5.67 \times 10^{-8} \times 4\pi \times (2 \times 10^{11})^2 \times (2000)^4$$

$$R_{\text{Total W.D}} = \sigma A_{WD} T_{WD}^4$$

$$= 5.67 \times 10^{-8} \times 4\pi \times (2 \times 10^6)^2 \times (20,000)^4$$

$$\approx 1.4\pi \times 10^{-8+12+16}$$

10. Ans (A,C)



$$\text{For ring, } ma_1 - f = ma_2$$

$$F - f = ma_1$$

$$\tau = fR = I\alpha$$

$$fR = mR^2 \frac{a_2}{R}, f = ma_2$$

$$a_1 = 2a_2, a_1 = \frac{2F}{3M}, a_2 = \frac{F}{3M}$$

PART-1 : PHYSICS

SECTION-II (i)

1. Ans (2.00)

Solid angle formed by a cone having semi vertical angle α is

$$\Omega = 2\pi(1 - \cos \alpha) = 2\pi(1 - \cos 60^\circ) = \pi \text{ Sr}$$

Therefore, one fourth of the total number of lines emitted from q_1 = Half the number of lines terminating on q_2

$$\therefore N_1 = 2N_2$$

$$\therefore |q_1| = 2|q_2|$$

2. **Ans (90.00)**

Half the lines emitted from q_1 do terminate on q_2 . Remaining half will go to ∞ .

$$\therefore \alpha_{\max} = 90^\circ$$

3. **Ans (18.00)**

Gas in A and B are always at equal temperature.

Process for gas in C is adiabatic.

$$\therefore P_0 V_0^{3/2} = P_C \left(\frac{4V_0}{9} \right)^{3/2} \therefore P_C = \frac{27}{8} P_0$$

For mechanical equilibrium $P_A = P_C$

$$\therefore P_A = P_C = \frac{27}{8} P_0$$

Again for C :

$$T_C \left(\frac{4V_0}{9} \right)^{3/2-1} = T_0 V_0^{3/2-1} \Rightarrow T_C = \frac{3}{2} T_0$$

For A : $P_0 V_0 = nRT_0$

$$\text{And } P_A \left[V_0 + \frac{5V_0}{9} \right] = nRT_A$$

$$[\because \text{volume change} = V_0 - \frac{4V_0}{9} = \frac{5V_0}{9}]$$

$$\therefore \frac{27}{8} P_0 \frac{14}{9} V_0 = nRT_A$$

$$\therefore T_A = \frac{21}{4} T_0$$

$$\text{Also } T_A = T_B = \frac{21}{4} T_0$$

Work done on gas in C = work done by gas in A.

$$W_A = - \left(\frac{P_0 V_0 - P_C V_C}{\gamma - 1} \right) = - \frac{P_0 V_0 - \frac{27}{8} P_0 \frac{4V_0}{9}}{0.5} = P_0 V_0$$

Work done by gas in B is zero.

Change in internal energy of A + B is

$$\Delta U_{A+B} = 2nC_V \Delta T = 2n \frac{R}{\gamma - 1} \left(\frac{21}{4} - 1 \right) T_0$$

$$= \frac{2nR}{0.5} \times \frac{17}{4} T_0 = 17nRT_0 = 17P_0 V_0$$

$$Q_{\text{Heater}} = \Delta U_{A+B} + W_{\text{by A}} = 18P_0 V_0$$

4. **Ans (9.50)**

Heat flow through piston 1

$$H = \Delta U_A + W_{\text{by A}}$$

$$= \frac{17}{2} P_0 V_0 + P_0 V_0 = \frac{19}{2} P_0 V_0$$

5. **Ans (0.10)**

$$\beta = \frac{\lambda f}{d} = \frac{5 \times 10^{-7} \times 20 \times 10^{-2}}{1 \times 10^{-3}} = 0.1 \text{ mm}$$

6. **Ans (0.40)**

$$n_1 B_1 = n_2 B_2$$

$$n_1 \times \frac{4000f}{d} = n_2 \times \frac{5000f}{d}$$

$$\Rightarrow \frac{n_1}{n_2} = \frac{5}{4} \text{ or } n_1 = 5 \text{ \& } n_2 = 4$$

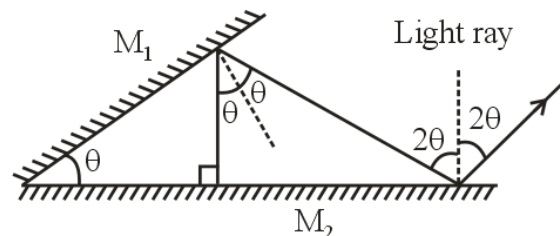
$$\Rightarrow \text{distance } 0.4 \text{ mm}$$

PART-1 : PHYSICS

SECTION-II (ii)

7. **Ans (9)**

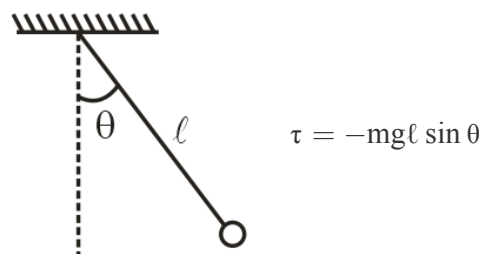
Ray starts retracing it's path when it falls normally on mirror. Let us do it in reverse order. 10th reflection must be on Mirror M_2 .



Angle of incidence after 10th reflection is θ , then 2θ . It increases in this order back to initial value of $90-\theta$.

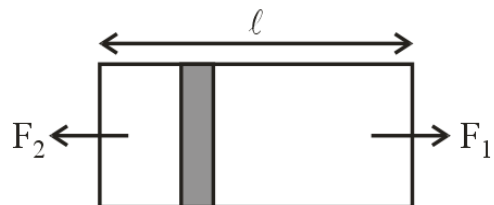
$$\text{Hence } 90 - \theta = 90 \Rightarrow \theta = 9$$

8. **Ans (3)**



Since $\theta < \sin \theta$, so Torque and consequent angular acceleration will be less, hence more time would be taken is oscillation.

9. Ans (2)



$$\int_{d\ell}^{\Delta\ell} d\ell = \int_0^{\ell} \frac{T dx}{AY}, T = F_1 - (F_1 - F_2) \frac{x}{\ell}$$

$$\Delta\ell = 0.12\text{cm}, A = 1\text{mm}^2 = 10^{-6}\text{m}^2$$

$$F_1 = 100\text{N}, F_2 = 80\text{N}$$

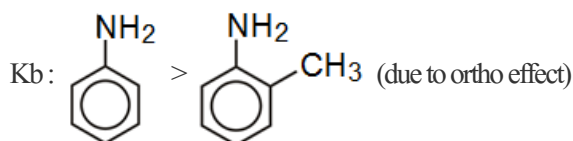
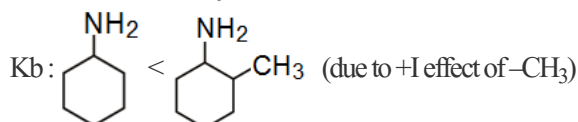
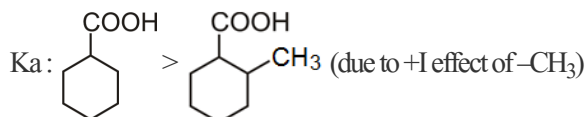
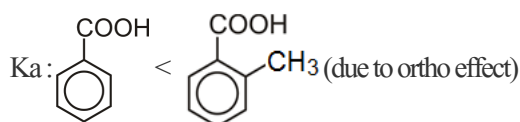
$$\Delta\ell = \frac{(F_1 + F_2) \ell}{2AY}, \ell = \frac{\Delta\ell (2AY)}{F_1 + F_2}$$

$$= \frac{0.12 \times 10^{-2} \times 2 \times 10^{-6} \times 1.5 \times 10^{11}}{180} = 2$$

PART-2 : CHEMISTRY

SECTION-I (i)

3. Ans (B)



PART-2 : CHEMISTRY

SECTION-I (ii)

7. Ans (B,D)

It is diels alder reaction

PART-2 : CHEMISTRY

SECTION-II (i)

1. Ans (10.00)

$$0.5 = \frac{1}{2} \log \left(\frac{V_2}{1} \right) \Rightarrow \frac{V_2}{1} = 10 \Rightarrow V_2 = 10\text{lit}$$

2. Ans (4.00)

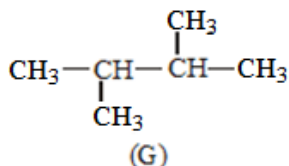
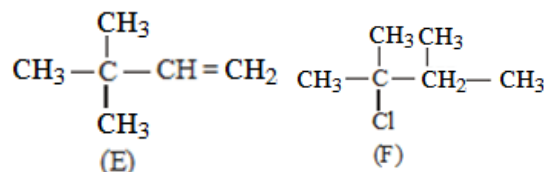
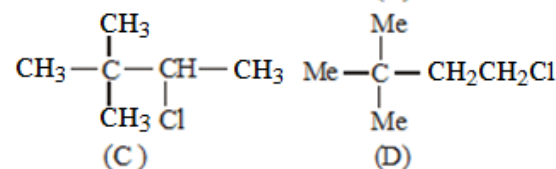
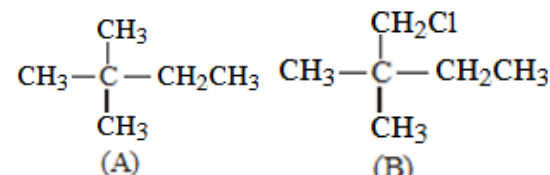
$$(\text{pH})_2 - (\text{pH})_1 = \frac{1}{2} \log \left(\frac{V_2}{V_1} \right)$$

$$\Rightarrow -\log \left(\frac{[\text{H}^+]}{2} \right) + \log [\text{H}^+] = \frac{1}{2} \log \left(\frac{V_2}{V_1} \right)$$

$$\Rightarrow \log 4 = \log \left(\frac{V_2}{V_1} \right)$$

$$\Rightarrow V_2 = 4\text{lit}$$

3. Ans (4.00)



PART-2 : CHEMISTRY

SECTION-II (ii)

7. Ans (8)

$$A \rightarrow 3B$$

$$1 - x \quad 3x$$

$$\text{at } t = 4, 1 - x = 3x$$

$$x = \frac{1}{4}$$

for zero order reaction

$$C_1 = C_0 - kt$$

$$\frac{3}{4} = 1 - k \times 4$$

$$4k = \frac{1}{4}; k = \frac{1}{16} \text{ and } t_{1/2} = \frac{C_0}{2k}$$

$$t_{1/2} = \frac{1}{2 \times \frac{1}{16}} = 8$$

PART-3 : MATHEMATICS

SECTION-I (i)

1. **Ans (D)**

$$\begin{aligned} AM &= \frac{A_6 + A_7}{2} = \frac{\alpha^6 + \beta^6 + \alpha^7 + \beta^7}{2} \\ &= \frac{\alpha^6(1 + \alpha) + \beta^6(1 + \beta)}{2} \\ &= \frac{-\alpha^8 - \beta^8}{2} = -\frac{A_8}{2} \end{aligned}$$

2. **Ans (C)**

$$\begin{aligned} T_n &= \frac{2n+3}{n(n+1)3^n} = \frac{3n+3-n}{n(n+1)3^n} \\ T_n &= \frac{1}{n \cdot 3^{n-1}} - \frac{1}{(n+1)3^n} \\ S_n &= \sum_{k=1}^n T_k = 1 - \frac{1}{(n+1)3^n} \\ \Rightarrow S_3 &= \frac{107}{108}, S_\infty = 1 \end{aligned}$$

3. **Ans (B)**

$$166320 = 2^4 \cdot 3^3 \cdot 7 \cdot 5 \cdot 11$$

$$792 = 2^3 \cdot 3^2 \cdot 11$$

$$a = 2^{x_1} 3^{y_1} 7^{z_1} 5^{u_1} 11^{w_1}$$

$$b = 2^{x_2} 3^{y_2} 7^{z_2} 5^{u_2} 11^{w_2}$$

$$\left. \begin{aligned} (x_1, x_2) &\rightarrow 2 \text{ ways} \\ (y_1, y_2) &\rightarrow 2 \text{ ways} \\ (z_1, z_2) &\rightarrow 2 \text{ ways} \\ (u_1, u_2) &\rightarrow 2 \text{ ways} \\ (w_1, w_2) &\rightarrow 1 \text{ way} \end{aligned} \right\} 16 \text{ ordered pair}$$

4. **Ans (A)**

$$P\left(\frac{R}{A}\right) = \left(\frac{2}{3}\right)^n \cdot P\left(\frac{R}{B}\right) = \left(\frac{1}{3}\right)^n, \text{ in } n \text{ throws}$$

$$P(R) = \frac{1}{2} \left(\frac{2}{3}\right)^n + \frac{1}{2} \left(\frac{1}{3}\right)^n, \text{ in } n \text{ throws}$$

$$\begin{aligned} P\left(\frac{A}{R}\right) &= \frac{P\left(\frac{R}{A}\right) P(A)}{P(R)} = \frac{\frac{1}{2} \left(\frac{2}{3}\right)^n}{\frac{1}{2} \left(\frac{2}{3}\right)^n + \frac{1}{2} \left(\frac{1}{3}\right)^n} \\ &= \frac{2^n}{2^n + 1} = \frac{32}{33} \Rightarrow n = 5 \end{aligned}$$

PART-3 : MATHEMATICS

SECTION-I (ii)

5. **Ans (A,B,C,D)**

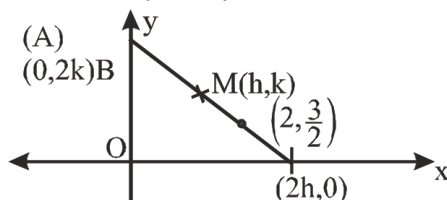
(A) Line AB is

$$\frac{x}{2h} + \frac{y}{2k} = 1. \text{ As } \left(2, \frac{3}{2}\right) \text{ lies on it}$$

$$\Rightarrow 3h + 4k = 4hk$$

$\therefore$ Locus of (h,k) is

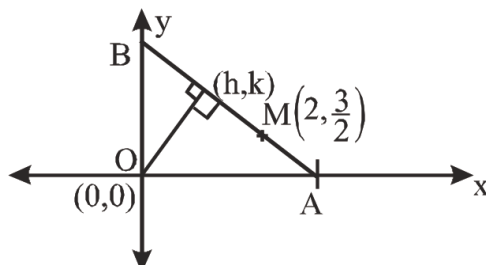
$$3x + 4y = 4xy.$$



(B) Line AB is

$$hx + ky = h^2 + k^2. \text{ As } \left(2, \frac{3}{2}\right) \text{ lies on it}$$

$$\Rightarrow 2h + \frac{3k}{2} = h^2 + k^2$$



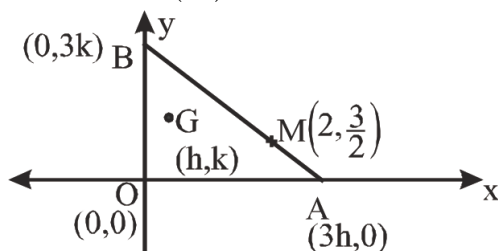
$\therefore$ Locus of (h,k) is

$$2(x^2 + y^2) - 4x - 3y = 0$$

(C) Line AB is

$$\frac{x}{3h} + \frac{y}{3k} = 1. \text{ As } \left(2, \frac{3}{2}\right) \text{ lies on it}$$

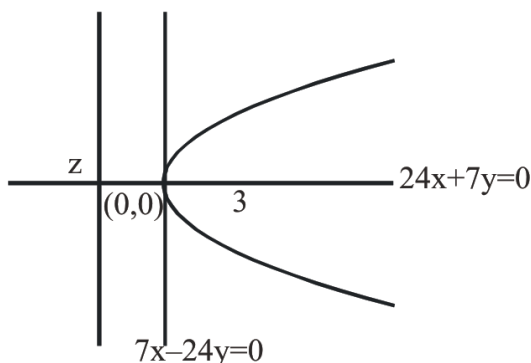
$$\Rightarrow \frac{2}{3h} + \frac{3}{3(2k)} = 1$$



$\therefore$ Locus of (h,k) is $3x + 4y = 6xy.$

6. Ans (A,C,D)

$$(24x+7y)^2 = 25(7x-24y)$$



$$\left(\frac{24x+7y}{\sqrt{24^2+7^2}} \right)^2 = \frac{(7x-24y)}{\sqrt{7^2+24^2}} \quad \text{LR} = 1$$

$$\vec{OS} = \frac{1}{4} \left(\frac{7\hat{i} - 24\hat{j}}{25} \right) \Rightarrow S \left(\frac{7}{100}, -\frac{24}{100} \right)$$

$$\vec{OZ} = -\frac{1}{4} \left(\frac{7\hat{i} - 24\hat{j}}{25} \right) \Rightarrow Z \left(-\frac{7}{100}, \frac{24}{100} \right)$$

7. Ans (A,B,C,D)

$$\sec^2 \alpha + 6\cos^2 \alpha = p \quad \dots(1)$$

$$\tan^2 \alpha - 5\sin \alpha = p - 8 \quad \dots(2)$$

$$(1) - (2) \Rightarrow 6 - 6\sin^2 \alpha - 5\sin \alpha = 7$$

$$\Rightarrow (2\sin \alpha - 1)(3\sin \alpha - 1) = 0$$

$$\Rightarrow \sin \alpha = \frac{1}{2} \text{ or } \sin \alpha = \frac{1}{3}$$

8. Ans (A,C,D)

As the equation does not change if x is replaced by $\frac{1}{x}$ so we can say the coefficients equidistant from ends are equal.

9. Ans (A,B,D)

ΔPQR is an equilateral triangle and its centroid lies on the circle $|z| = 1$

$$\text{Also, } \left| \frac{z_1 + z_2 + z_3}{3} - z_3 \right| = 1$$

10. Ans (B,C)

Let $P(4\cos\theta, 3\sin\theta)$

$\therefore r + s$ has minimum integral value -5

Chord of contact of C_2 is

$$2x - 3y\sin\theta + 8\cos\theta = 0 \quad \dots\dots(i)$$

Let mid point (h, k) , then chord of contact is $2x$

$$-ky + k^2 - 2h = 0 \quad \dots\dots(ii)$$

(i) & (ii) same

$$k = 3\sin\theta \text{ \& } k^2 - 2h = 8\cos\theta$$

$$\therefore \text{ lie on } \frac{y^2}{9} + \frac{(y^2 - 2x)^2}{64} = 1$$

PART-3 : MATHEMATICS

SECTION-II (i)

1. Ans (8.00)

Let $f(x) = ax + b$

Since f is increasing then

$$f(-1) = 3 \Rightarrow -a + b = 3$$

$$f(1) = 5 \Rightarrow a + b = 5$$

hence $f(x) = x + 4$

$$f^{-1}(x) = x - 4$$

$$f^{-1} \left(\frac{15}{4} \right) = \frac{15}{4} - 4 = -\frac{1}{4}$$

2. Ans (-0.25)

Let $f(x) = ax + b$

Since f is increasing then

$$f(-1) = 3 \Rightarrow -a + b = 3$$

$$f(1) = 5 \Rightarrow a + b = 5$$

hence $f(x) = x + 4$

$$f^{-1}(x) = x - 4$$

$$f^{-1} \left(\frac{15}{4} \right) = \frac{15}{4} - 4 = -\frac{1}{4}$$

3. Ans (1.00)

$$\int \frac{f'(x) - f(x)}{f(x) - f'(x)} dx = \int dx$$

$$f(x) - f'(x) = \lambda e^x.$$

$$e^{-x} f(x) = -\lambda x + c$$

$$f(x) = -\lambda x e^x + ce^x$$

put $x = 0$ & $x = 1$ to get $c = 0$ & $1 = -3$.

$f(x) = 3x e^x$. $\Rightarrow$ the equation $3x e^x = 0$ has only one solution i.e. $x = 0$.

4. Ans (-1.00)

$$\begin{aligned} & \int_0^1 \left(\frac{e^x \cdot x^3}{2} - e^x (3x) \right) dx \\ &= \int_0^1 \left(e^x \frac{x^3}{2} + e^x \frac{3x^2}{2} - e^x \frac{3x^2}{2} - e^x (3x) \right) dx \\ &= \int_0^1 e^x \left(\frac{x^3}{2} + \frac{3x^2}{2} \right) dx - \int_0^1 e^x \left(\frac{3x^2}{2} + 3x \right) dx \\ &= \left(e^x \cdot \frac{x^3}{2} \right) \Big|_0^1 - \left(e^x \frac{3x^2}{2} \right) \Big|_0^1 \Rightarrow \frac{e}{2} - \frac{3e}{2} = -e. \end{aligned}$$

5. Ans (1.00)

$$\begin{aligned} f'(x) &= x^2(x-2)(6x^2-13x+6) \\ &= x^2(x-2)(3x-2)(2x-3) \end{aligned}$$

$\begin{array}{ccccccc} - & - & + & - & + & & \\ 0 & \frac{2}{3} & \frac{3}{2} & 2 & & & \end{array}$

6. Ans (1.00)

$$\begin{aligned} f'(x) &= x^2(x-2)(6x^2-13x+6) \\ &= x^2(x-2)(3x-2)(2x-3) \end{aligned}$$

$\begin{array}{ccccccc} - & - & + & - & + & & \\ 0 & \frac{2}{3} & \frac{3}{2} & 2 & & & \end{array}$

PART-3 : MATHEMATICS

SECTION-II (ii)

7. Ans (6)

$$\begin{aligned} \frac{a_{n+1}}{a_n} &= \frac{n+4}{n+1} \\ \frac{a_2}{a_1} &= \frac{5}{2}, \frac{a_3}{a_2} = \frac{6}{3}, \frac{a_4}{a_3} = \frac{7}{4} \dots\dots \\ \therefore \frac{a_2}{a_1} \times \frac{a_3}{a_2} \times \frac{a_4}{a_3} \times \dots \times \frac{a_{n+1}}{a_n} \\ &= \frac{5}{2} \times \frac{6}{3} \times \frac{7}{4} \times \frac{8}{5} \times \frac{9}{6} \times \frac{10}{7} \times \dots \times \frac{n+3}{n} \times \frac{n+4}{n+1} \\ a_n &= \frac{(n+1)(n+2)(n+3)}{2} \\ \therefore \frac{1}{a_n} &= \frac{2}{(n+1)(n+2)(n+3)} \\ &= \frac{1}{n+1} - \frac{2}{n+2} + \frac{1}{n+3} \\ \lim_{r \rightarrow \infty} \sum_{k=1}^r \frac{1}{a_k} \\ &= \lim_{r \rightarrow \infty} \left[\left(\frac{1}{2} - \frac{2}{3} + \frac{1}{4} \right) + \left(\frac{1}{3} - \frac{2}{4} + \frac{1}{5} \right) + \dots \right] \\ &= \lim_{r \rightarrow \infty} \left(\frac{1}{6} - \frac{1}{(r+2)(r+3)} \right) \\ &= \lim_{r \rightarrow \infty} \left(\frac{r^2 + 5r}{6r^2 + 30r + 36} \right) = \frac{1}{6} \end{aligned}$$

8. Ans (2)

$$\begin{aligned} x^2 - 4|x| + 6 &= 2\cos(\pi x) \\ \Rightarrow (|x| - 2)2 + 2 &= 2\cos(\pi x) \\ \Rightarrow x &= \pm 2 \\ k &= \int_{-2}^2 [x^2 + 6 - 4|x| - 2\cos(\pi x)] dx \\ &= 2 \int_0^2 (x^2 - 4x + 6 - 2\cos \pi x) dx \end{aligned}$$

9. Ans (6)

$$\begin{aligned} k &= \int_{-2013}^{2013} \frac{1}{(1+x^{2011} + \sqrt{1+x^{4022}})(1+x^{2011} - \sqrt{1+x^{4022}})} \cdot dx \\ &= \int_{-2013}^{2013} \frac{1+x^{2011} - \sqrt{1+x^{4022}}}{2 \cdot x^{2011}} \cdot dx \\ &= \int_{-2013}^{2013} \underbrace{\frac{1}{2 \cdot x^{2011}}}_{\text{odd function}} dx + \int_{-2013}^{2013} \frac{1}{2} \cdot dx - \int_{-2013}^{2013} \underbrace{\frac{\sqrt{1+x^{4022}}}{x^{2011}}}_{\text{odd function}} dx \\ &= \int_{-2013}^{2013} \underbrace{\frac{1}{2 \cdot x^{2011}}}_{\text{odd function}} dx + \int_{-2013}^{2013} \frac{1}{2} \cdot dx - \int_{-2013}^{2013} \underbrace{\frac{\sqrt{1+x^{4022}}}{x^{2011}}}_{\text{odd function}} dx \end{aligned}$$

JEE(Main + Advanced) : ENTHUSIAST COURSE : [SCORE-II (ALL STAR BATCH)]

ANSWER KEY

PAPER-2

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	B,C	B	A,C	A,C	A,D	A,C
SECTION-I (ii)	Q.	7	8	9	10		
	A.	A	C	C	B		
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	7.00	100.00	0.25	4.00	178.89	709.56
SECTION-II (ii)	Q.	7	8	9			
	A.	10	8	7			

PART-2 : CHEMISTRY

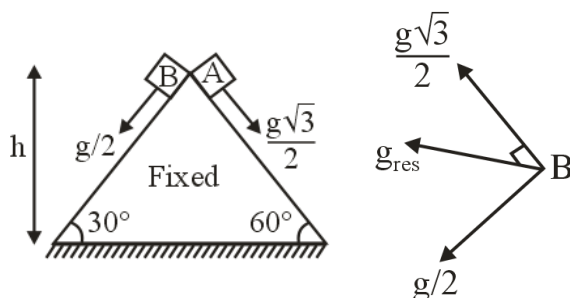
SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	A,B,C	B,C,D	A,B,C,D	B,C	B,C	A,B
SECTION-I (ii)	Q.	7	8	9	10		
	A.	C	C	C	B		
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	228.00	46.00	5.00	3.00	3.00	3.00
SECTION-II (ii)	Q.	7	8	9			
	A.	5	46	5			

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	B,C,D	A,C	A,B,C,D	A,B,C	B,C	A,B,C,D
SECTION-I (ii)	Q.	7	8	9	10		
	A.	C	B	A	B		
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	20.00	6.00	162.00	1944.00	3.00	1.00
SECTION-II (ii)	Q.	7	8	9			
	A.	3	7	4			

HINT – SHEET
PART-1 : PHYSICS
SECTION-I (i)

1. Ans (B,C)



g resultant of B with respect to A is

$$= \sqrt{\left(\frac{g}{2}\right)^2 + \left(\frac{g\sqrt{3}}{2}\right)^2} = \sqrt{g^2 \left[\frac{1}{4} + \frac{3}{4}\right]} = g$$

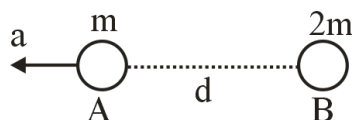
By energy conservation,

$$\text{For B } m_B gh = \frac{1}{2} m v_B^2$$

$$\text{For A } m_A gh = \frac{1}{2} m v_A^2$$

$$t = \sqrt{\frac{2h}{g \sin^2 \theta}}, \text{ so time taken is different.}$$

2. Ans (B)



Due to attraction speed of A decreases and than of B increases, to escape $V_A > V_B$ at infinite separation, for minimum value of v , $V_A = V_B$ at infinite separation.

$$mv = 3v_f \Rightarrow v_f = \frac{v}{3}$$

$$\text{now } \frac{1}{2}mv^2 - 2\frac{am^2}{d} = \frac{1}{2}(m+2m)\left(\frac{v}{3}\right)^2$$

$$v = \sqrt{\frac{6GM}{d}}$$

3. Ans (A,C)

$$(A) V_{AB} = \frac{\epsilon}{\left(\frac{6\rho l}{\pi a^2}\right)} \left(\frac{5\rho l}{\pi a^2}\right)$$

$$(C) V_{AB} = \frac{5\epsilon}{6}$$

$$\epsilon' = (i) \left(\frac{\rho l}{\pi a^2} + \frac{2\rho l}{\pi a^2}\right) = (i) \frac{\epsilon}{\left(\frac{6\rho l}{\pi a^2}\right)} \cdot \frac{3\rho l}{\pi a^2} = \frac{\epsilon}{2}$$

4. Ans (A,C)

$$\frac{\lambda_0}{\lambda_1} = 4 \Rightarrow \frac{(Z_1 - 1)^2}{(Z - 1)^2} = 4 \Rightarrow Z_1 = 2Z - 1$$

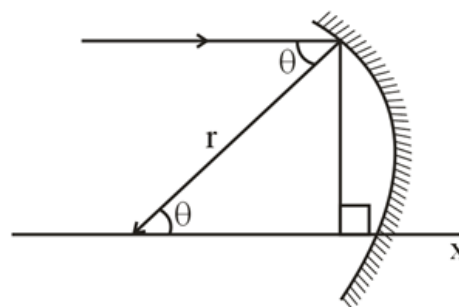
$$\frac{\lambda_0}{\lambda_2} = \frac{1}{4} \Rightarrow \frac{(Z_2 - 1)^2}{(Z - 1)^2} = \frac{1}{4} \Rightarrow Z_2 = \frac{Z + 1}{2}$$

5. Ans (A,D)

Shift in fringe pattern

$$\begin{aligned} &= \frac{t_2(\mu - 1)}{\lambda} \beta - \frac{t_1(\mu - 1)}{\lambda} \beta = \frac{(t_2 - t_1)(\mu - 1)\beta}{\lambda} \\ &= \frac{d \tan \theta (\mu - 1)\beta}{\lambda} \text{ const } \tan \theta; = \frac{2t(\mu - 1)D}{3d} \end{aligned}$$

6. Ans (A,C)



$$d = \sqrt{1+3} = 2$$

$$r = 4$$

$$\sin \theta = \frac{1}{2} \Rightarrow \theta = 30^\circ$$

$$\text{now CF} = \frac{R}{2} \sec \theta = \frac{4}{\sqrt{3}}$$

$$\angle = 180 - 20 = 120^\circ$$

PART-1 : PHYSICS

SECTION-I (ii)

7. Ans (A)

$$\begin{aligned} mgh &= \frac{1}{2}mv^2 + \frac{1}{2}mv^2 + \frac{1}{2} \cdot \frac{2}{5}mr^2 \left(\frac{2v}{r}\right)^2 \\ &= \frac{1}{2}mv^2 \left(1 + 1 + \frac{8}{5}\right) = \frac{1}{2}mv^2 \frac{18}{5} = \frac{9mv^2}{5} \end{aligned}$$

$$\Rightarrow v = \sqrt{\frac{5}{9}gh}$$

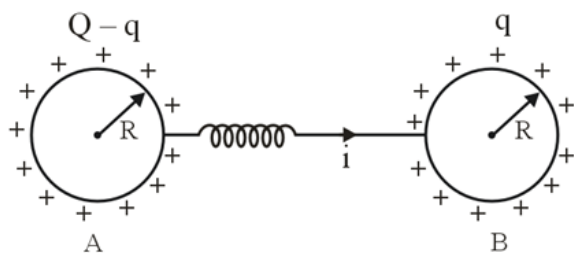
8. Ans (C)

$$\begin{aligned} \text{K.E. of ball} &= \frac{1}{2}mv^2 + \frac{1}{2} \left(\frac{2}{5}mr^2\right) \left(\frac{2v}{r}\right)^2 + mgh \\ &= \frac{mv^2}{2} + \frac{4}{5}mv^2 + mgh = \frac{31}{18}mgh \end{aligned}$$

10. Ans (B)

For Q. 9 & 10

Let q = charge on B at time ' t '



$$V_A - V_B = L \frac{di}{dt} \Rightarrow K \frac{Q-q}{R} - K \frac{q}{R} = L \frac{d^2q}{dt^2}$$

[Spheres are at large distance. Hence charge on one does not affect the potential of other]

$$\Rightarrow \frac{d^2q}{dt^2} = -\frac{2K}{LR} \left[q - \frac{Q}{2} \right] \dots (i)$$

$$\text{Let } q - \frac{Q}{2} = x \text{ which means } \frac{d^2q}{dt^2} = \frac{d^2x}{dt^2}$$

In terms of x , the differential equation (i) becomes

$$\frac{d^2x}{dt^2} = -\frac{2K}{LR} x$$

Solution to this equation is

$$x = x_0 \sin(\omega t + \delta) \left[\omega = \sqrt{\frac{2K}{LR}} \right]$$

$$\Rightarrow q - \frac{Q}{2} = x_0 \sin(\omega t + \delta)$$

$$\Rightarrow q = \frac{Q}{2} + x_0 \sin(\omega t + \delta)$$

$$\text{Current } i = \frac{dq}{dt} = x_0 \omega \cos(\omega t + \delta)$$

Just after closing the switch (at $t = 0$) the current is zero

$$\therefore \delta = \frac{\pi}{2} \therefore q = \frac{Q}{2} + x_0 \cos \omega t$$

Also at $t = 0$, $q = 0$

$$\therefore x_0 = -Q/2, q = \frac{Q}{2} [1 - \cos \omega t]$$

$$q = \frac{Q}{2} \text{ means } \cos \omega t = 0$$

$$\Rightarrow \omega t = \frac{\pi}{2} \Rightarrow \sqrt{\frac{2K}{LR}} t = \frac{\pi}{2} \Rightarrow t = \frac{\pi}{2} \sqrt{\frac{LR}{2K}}$$

PART-1 : PHYSICS

SECTION-II (i)

2. Ans (100.00)

$$\frac{x^2}{16} + \frac{y^2}{7} = 1 \Rightarrow e = \sqrt{1 - \frac{7}{16}} = \frac{3}{4}$$

$$v_{\max} = \sqrt{\frac{GM}{a} \left(\frac{1+e}{1-e} \right)}, v_{\min} = \sqrt{\frac{GM}{a} \left(\frac{1-e}{1+e} \right)}$$

$$\text{So, } \frac{v_{\max}}{v_{\min}} = \frac{1+e}{1-e} = 7$$

at Aphelion speed is minimum, for

$$\text{circular orbit } v = \sqrt{\frac{GM}{r}} = \sqrt{\frac{GM}{a(1+e)}}$$

($r = a(1+e)$, as Aphelion)

$$\text{so, } v = \frac{v_{\min}}{\sqrt{1-e}}$$

$$\text{percentage change} = \left(\frac{1}{\sqrt{1-e}} - 1 \right) \times 100 = 100\%$$

3. Ans (0.25)

$$\frac{X_i}{X_f} = \left| \frac{\frac{1}{\omega C_i} - \omega L_i}{\frac{1}{\omega C_f} - \omega L_f} \right|$$

$$\frac{\frac{\ell}{\omega K \epsilon_0 A} - \omega \mu_0 n^2 A \ell}{\frac{\ell}{\omega \epsilon_0 A} - \omega \mu_0 \mu_r n^2 A \ell} \cdot \frac{1}{K} \left(\frac{1-K}{1-\mu_r} \right)$$

$$[\text{Using } \epsilon_0 \mu_0 = \frac{1}{C^2}] \text{ \& } [\omega^2 A^2 n^2 = C^2]$$

4. Ans (4.00)

$$C_1 = \frac{\epsilon_0 A/2}{\ell}, C_2 = \frac{K \epsilon_0 A/2}{\ell}$$

$C + C_1 + C_2$ (parallel combination)

$$C = \frac{\epsilon_0 A}{2\ell} (1 + K)$$

$$L_1 = \mu_0 n^2 A \frac{\ell}{2}$$

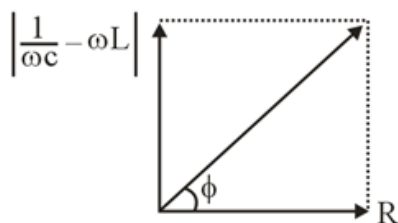
$$L_2 = \mu_0 \mu_r n^2 \frac{A\ell}{2}$$

$$L = L_1 + L_2$$

(series combination)

$$L = \mu_0 n^2 A \frac{\ell}{2} (1 + \mu_r)$$

Now power factor



$$\cos \phi = \frac{1}{\sqrt{2}}$$

$$\Rightarrow \phi = 45^\circ$$

$$\text{So, } R = \left| \frac{1}{\omega C} - \omega L \right|$$

On solving we get are equal to $\frac{4\ell}{3\epsilon_0 A \omega}$

5. Ans (178.89)

Distance travelled by car in 1 sec

$$\frac{500}{18} = 27.78 \text{ m}$$

Distance travelled by sound in 1 sec = 330 m

so initial distance between car and cliff

$$= \frac{330 + 27.78}{2} \Rightarrow 178.89 \text{ m}$$

6. Ans (709.56)

$$f_{\text{act}} = \frac{v - v_c}{v + v_c} f_{\text{app}}$$

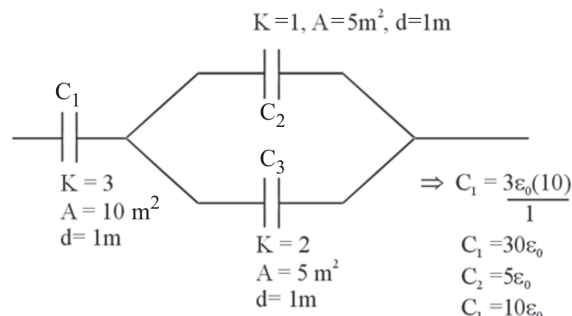
$$= \left(\frac{330 - \frac{500}{18}}{330 + \frac{500}{18}} \right) \times 840 = 709.56 \text{ Hz}$$

PART-1 : PHYSICS

SECTION-II (ii)

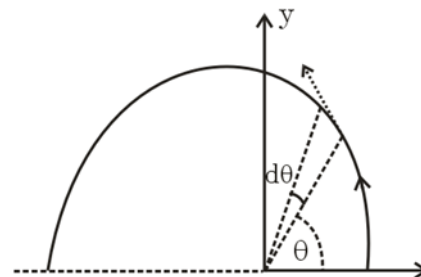
7. Ans (10)

Finally system behaves as shown



$$(\text{net}) = \frac{(15\epsilon_0)(30\epsilon_0)}{45\epsilon_0} = 10\epsilon_0 = 8.85 \times 10^{-11} \text{ F}$$

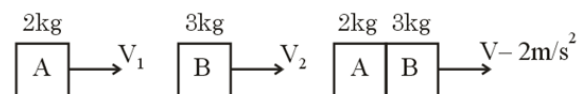
8. Ans (8)



$$d\vec{B} = \frac{\mu_0 I}{4\pi} \frac{d\vec{l} \times \hat{r}}{r^2}$$

$$dB = \frac{\mu_0 I}{4\pi} \frac{rd\theta}{r^2}$$

9. Ans (7)



$$2(V_1) + 3(V_2) = (2 + 3)V$$

$$\Rightarrow 2V_1 + 3V_2 = 20 \text{ ..(i)}$$

loss in KE = heat energy

$$\frac{1}{2} 2(V_1)^2 + \frac{1}{2} 3(V_2)^2 - \frac{1}{2} (2 + 3)V^2 = (2 + 3)S\Delta T$$

$$V_1^2 + \frac{3}{2} V_2^2 - 40 = 15 \text{ or } V_1^2 + \frac{3V_2^2}{2} = 55 \text{ ... (ii)}$$

Solving equation (i) and (ii), we get,

$V_1 = 1 \text{ m/s}$ or 7 m/s and $V_2 = 6 \text{ m/s}$ or 2 m/s for collision $V_1 > V_2$.

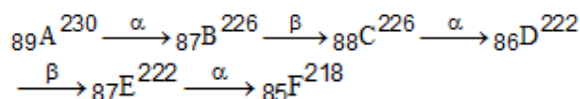
So $V_1 = 7 \text{ m/s}$ and $V_2 = 2 \text{ m/s}$

PART-2 : CHEMISTRY

SECTION-I (i)

2. Ans (B,C,D)

From the given option, complete the chain reaction



* D undergoes β^- decay

* Atomic number of A is 89 while E has 87

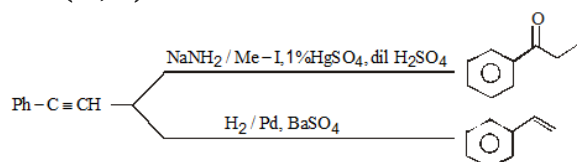
* Mass number of F is 218 while mass number of B is 226

* Number of neutron in C is 138 and correspondingly A is ${}_{89}A^{230}$

3. Ans (A,B,C,D)

A and D SIR effect

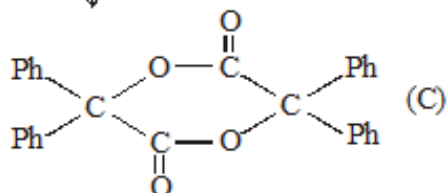
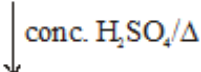
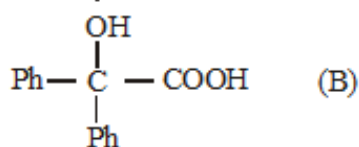
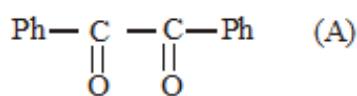
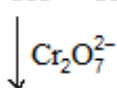
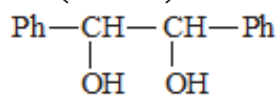
4. Ans (B,C)



PART-2 : CHEMISTRY

SECTION-II (i)

1. Ans (228.00)



PART-2 : CHEMISTRY

SECTION-II (ii)

8. Ans (46)

$\text{C}_5\text{H}_{10}\text{O}_5$ is pentose sugar

pentasaccharide means joining five molecules by glucosidic bond

$5(\text{C}_5\text{H}_{10}\text{O}_5) - 4(\text{H}_2\text{O})$

i.e. $\text{C}_{25}\text{H}_{42}\text{O}_{21}$

Hence $25 + 42 - 21 = 46$

PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (B,C,D)

$$D = \begin{vmatrix} 1 & 1 & a \\ 2 & 3 & 0 \\ 3 & 4 & a^2 \end{vmatrix} = a(a-1)$$

for $a = 0$ $D = 0$, $D_x = 0$, $D_y = 0$, $D_z = 2 - b$

so for $a = 0$, $b = 2 \Rightarrow$ Infinite Solutions.

for $a = 0$, $b = 1 \Rightarrow$ No solution.

for $a = 1$, $D = 0$, $D_x = 0$, $D_y = 0$, $D_z = 0$

$\Rightarrow$ Infinite solutions.

If $a \neq 0$ but $a = 1$ doesn't have unique solutions.

2. Ans (A,C)

Equation of family of planes

$$(2x - y + z - 2) + \lambda(x + 2y - z - 3) = 0$$

It must satisfy (3, 2, 1)

$$(6 - 2 + 1 - 2) + \lambda(3 + 4 - 1 - 3) = 0$$

$$3 + 3\lambda = 0 \Rightarrow \lambda = -1$$

$$x - 3y + 2z + 1 = 0$$

Equation of acute angle bisector

$$\frac{2x - y + z - 2}{\sqrt{6}} = \frac{x + 2y - z - 3}{\sqrt{6}}$$

$$x - 3y + 2z + 1 = 0$$

3. Ans (A,B,C,D)

Given, $a_1 + a_2 (1 - 2 \sin^2 x) + a_3 \sin^2 x = 0$
 $\forall x \in \mathbb{R}$

$$\Rightarrow (a_1 + a_2) + (a_3 - 2a_2) \sin^2 x = 0$$

So, $a_1 + a_2 = 0$ and $a_3 = 2a_2$ must hold simultaneously.

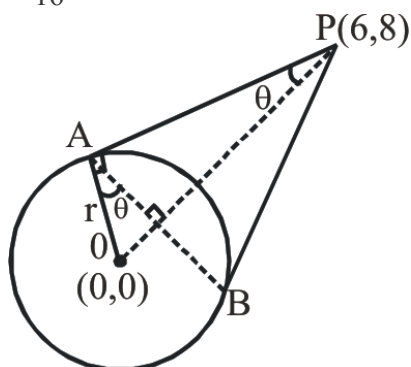
$$\text{So, } \frac{a_1}{-1} = \frac{a_2}{1} = \frac{a_3}{2} = \lambda (\neq 0)$$

$$\Rightarrow (a_1, a_2, a_3) = (-\lambda, \lambda, 2\lambda) \forall \lambda \in \mathbb{R}_0.$$

4. **Ans (A,B,C)**

$$\text{Area of } \Delta PAB = 100 \sin \theta \cos^3 \theta$$

$$\frac{3\sqrt{3}}{16} = \sin \theta \cdot \cos^3 \theta$$



$$\therefore \theta = \frac{\pi}{6} \Rightarrow r = 5$$

Equation of chord of contact is $T = 0$

$$6x + 8y = r^2$$

OP is diameter of circumcircle of DABP

$$\therefore \text{equation } x^2 + y^2 - 6x - 8y = 0$$

5. **Ans (B,C)**

$$\text{As } \frac{z_1 + z_2}{2} = 3 + 2i$$

$$\text{and } \frac{\bar{z}_1 + \bar{z}_2}{2} = 3 - 2i$$

$$t = 12$$

Now verify options.

6. **Ans (A,B,C,D)**

$$AA^T = I = A^T A \Rightarrow |A| = \pm 1 \text{ similarly for B}$$

$$\text{Let } C = ABA \Rightarrow CC^T = ABA \cdot A^T B^T A^T = I$$

$$D = A^2 B^2 \Rightarrow DD^T = AABBB^T B^T A^T A^T = I$$

&

$$\left| \frac{A}{2} \text{adj}(2B) \right| = |2A \text{adj}(B)| = 8|A||\text{adj}B| = 8|A||B|^2$$

PART-3 : MATHEMATICS

SECTION-I (ii)

7. **Ans (C)**

$$\text{Weight of red coming at any place} = \frac{2}{9}$$

8. **Ans (B)**

Since red came at 4th place so weight of black coming at any place = $\frac{3}{8}$

9. **Ans (A)**

Figure is a square of side $2\sqrt{2}a$.

$$\therefore \text{Area} = (2\sqrt{2}a)^2 = 8 \quad (\because a = 1)$$

10. **Ans (B)**

Centre is focus of parabola = (1, 0)

PART-3 : MATHEMATICS

SECTION-II (i)

1. **Ans (20.00)**

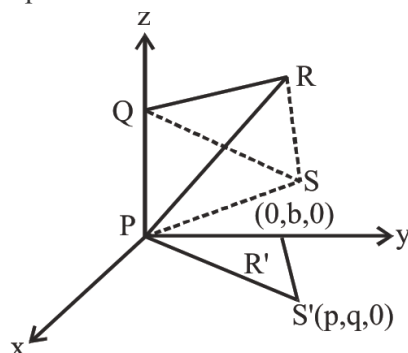
$$P(0, 0, 0); Q(0, 0, 4)$$

$$R(0, b, c); S(p, q, r)$$

P is same as P'

$$\text{Ar}(\Delta PRS') = \frac{1}{2}bp = 15$$

$$bp = 30$$



$$\text{Vol}(PQRS) = \frac{1}{3} \times (\text{Ar}PQR) \times \text{height}$$

$$= \frac{1}{2} \times \left(\frac{1}{2} \times 4 \times b \right) \times p$$

$$= \frac{1}{3} \times \frac{1}{2} \times 4 \times 30 = 20 \quad \text{cu. units.}$$

$$\text{if } P = 5, b = 6$$

3. **Ans (162.00)**

Let a, b, c & d are a, ar, ar², (2ar² - ar)

$$\therefore a + 2ar^2 - ar = 66 \text{ \& } ar + ar^2 = 60$$

$$\Rightarrow \frac{1 + 2r^2 - r}{r + r^2} = \frac{66}{60}$$

$$\Rightarrow r = \frac{2}{3} \text{ or } \frac{5}{3} \text{ (reject)}$$

$$\Rightarrow a = 54$$

GP is 54, 36, 24,

AP is 36, 24, 12, 0,

$$\text{Sum of infinite G.P.} = \frac{54}{1 - \frac{2}{3}} = 162$$

4. **Ans (1944.00)**

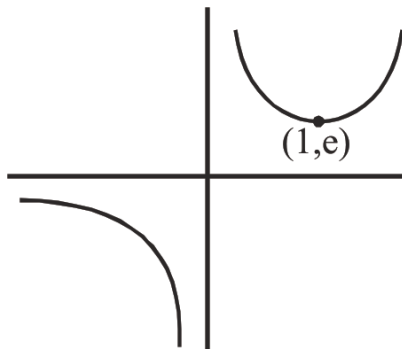
$$\sum_{i=1}^{\infty} U_i T_i = \text{Sum of } \infty \text{ A.G.P.}$$

$$= \frac{ab}{1-r} + \frac{adr}{(1-r)^2}$$

$$S = \frac{54 \times 36}{1 - \frac{2}{3}} + \frac{54(-12) \times \frac{2}{3}}{(1 - \frac{2}{3})^2}$$

$$= 54 \times 3(36 - 24) = 1944$$

5. **Ans (3.00)**



$$k \in (e, \infty)$$

6. **Ans (1.00)**

$$F(x) = e^x + \frac{e^x \cdot x - e^x}{x^2}$$

$$= \frac{(x^2 + x - 1)e^x}{x^2} > 0$$

PART-3 : MATHEMATICS

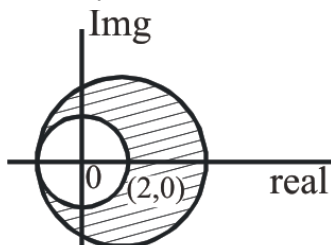
SECTION-II (ii)

7. **Ans (3)**

$$z + \frac{1}{z} = (x + iy) + \frac{(x - iy)}{x^2 + y^2}$$

$$\text{Im}g \left(z + \frac{1}{z} \right) = 0 \Rightarrow y - \frac{y}{x^2 + y^2} = 0$$

$$\Rightarrow x^2 + y^2 = 1$$



$$\text{Area} = \pi(2)^2 - \pi(1)^2 = 3\pi^2$$

8. **Ans (7)**

$$\text{Let } f(x) = (1 - x^{2021})^{\frac{1}{2022}}$$

$$\& g(x) = (1 - x^{2022})^{\frac{1}{2021}}$$

clearly $f(x)$ and $g(x)$ are inverse of each other also,

$$\int_0^1 f(x) dx + \int_1^0 g(x) dx = 0 \Rightarrow \int_0^1 f(x) dx = \int_0^1 g(x) dx$$

$$\text{i.e. } I_1 = I_2$$

$$\Rightarrow \frac{3I_1 + 4I_2}{2I_1 - I_2} = 7$$

9. **Ans (4)**

$$\text{I } 999986 \rightarrow 2 \frac{5!}{4!} = 10$$

$$\text{II } 999887 \rightarrow \frac{5!}{3!} = 20$$

$$\text{III } 998888 \rightarrow \frac{5!}{2!3!} = 10$$

$$\text{Total } 40$$